Scraping Quality Benchmark

Home Assignment

Purpose

Tavily provides agents with the most relevant information from the web at speed. Our system balances three trade-offs: **latency**, **accuracy**, and **cost**.

In this assignment, you'll build a web scraper and benchmark it against our production system on content extraction quality. You'll receive URLs with ground truth, optimize your approach, then run it on a held-out test set.

Your Task

Build a scraping system that:

1. Scrapes 500 training URLs (mix of static and JS-heavy sites) and measures quality using the provided scoring script.
2. Iterates and improves your approach using the training set feedback.
3. Runs your final scraper on 500 test URLs and submits the results.
4. Handles dynamic rendering and CAPTCHAs where possible.

What You Receive

File	Description
`train.csv`	500 URLs with ground truth (`truth_text` , `lie_text`) for development
`test.csv`	500 URLs (no ground truth) — your final evaluation set
`score.py`	Scoring script to evaluate your results on the training set
`proxy.json`	Proxy credentials (residential proxy) for you to use
`assignment.pdf`	This document

How Scoring Works

Each URL in the training set has:

- `truth_text` — important content that should appear in your scraped output.
- `lie_text` — boilerplate/navigation text that should ideally be excluded.

The scorer uses a **sliding window** over your extracted content to find the best-matching region against the truth text, then computes:

Metric	What It Measures
Success Rate	% of URLs with status 200-399, non-empty content, no block page detected
Precision	Of the tokens in the best window, how many match the truth?
Recall	Of the truth tokens, how many appear in the best window?
F1	Harmonic mean of precision and recall

Run the scorer on your training results:

```
python score.py --results my_results.jsonl --ground-truth train.csv
```

Submission Format

Submit your results as a **JSONL** file (one JSON object per line):

```
{
  "id": 1, "url": "https://...",
  "content": "extracted text...",
  "status_code": 200, "latency": 1.23
}
```

Field	Description
`id`	Task ID matching the CSV
`url`	The URL that was scraped
`content`	Extracted page content (**not** raw HTML)
`status_code`	HTTP status code
`latency`	Scrape time in seconds

You must submit **two JSONL files**: one for training results, one for test results.

Deliverables

1. **Python/Colab Notebook**
   a. Implement scraping with at least two approaches (lightweight fetch + JS-enabled browser).
   b. Show your iterative improvement process using the training set scores.
   c. Benchmark and visualize results (success rates, F1, latency, content quality).
   d. Link to a GitHub repository. Use a flow chart to show your code flow.
2. **Results Files**
   a. `train_results.jsonl` — your training set results (scored by you).
   b. `test_results.jsonl` — your test set results (scored by us).
3. **One-pager PDF**
   a. Outline your approach, trade-offs, and findings.
   b. Highlight limitations or challenges you faced.
   c. Include your training set scores and analysis.

Requirements

1. Some pages may present CAPTCHAs or bot-detection challenges. How you handle these is up to you — document your approach and trade-offs.
2. Notebook must be easy to follow (comments, headings, plots).
3. Support multilingual pages without breaking.

Evaluation Criteria

> **Your scraper will be benchmarked against Tavily's production system on the test set.**

1. **Reliability:** Scrape success rates, error handling.
2. **Quality:** F1, precision, and recall scores on the test set.
3. **Engineering:** Code structure, clarity, scalability.
4. **Insights:** Quality of analysis, iteration process, and commentary.
5. **Feasibility:** Realistic trade-offs for production (speed vs. cost).

Tips

- The dataset includes social media, PDFs, news, e-commerce, academic papers, and more. Different domains need different strategies.
- Some URLs are *expected* to fail (404s, paywalls). Don't chase 100%.
- Scraping is only half the job — extracting clean, meaningful content from raw HTML is what drives your quality score.
- Focus on the content that matters (`truth_text`) and filter out noise (`lie_text`).
- Use the proxy — many sites will block datacenter IPs.
- Show your thought process: what did you try, what improved scores, what didn't work?

Due Date

1 week from receiving this assignment.

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